

Quick Check

Lesson 6: Building Our Chemicals of Life Model: Consolidating the Science Behind Two Children in Kisumu

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.2: Chemicals of Life | Lesson 6

At the start of the unit, a group wrote: "Food is just energy, so two children eating the same amount should be equally healthy." Which updated statement shows the best evidence-based change in thinking?

- A** Only carbohydrates matter, because they give the body glucose.
- B** Food is made of six chemicals of life with different structures and functions, and missing one can cause a specific disease.
- C** Food is mainly energy, but children who eat more of it become healthier.
- D** Food with more water in it is healthier because water is a chemical of life.

On a group poster about two children in Kisumu, one child eats only ugali (maize starch) every day and the other eats githeri (maize and beans). Which arrow label correctly links the ugali child to the chemicals of life?

- A Excess protein and too little carbohydrate
- B Plenty of carbohydrate but too little protein
- C No carbohydrate at all, so no energy
- D Balanced intake of all six chemicals

A learner crushes avocado, shakes it with ethanol, then pours the liquid into water. A cloudy white layer forms. What does this ethanol emulsion result show?

- A** Starch is present
- B** Glucose is present
- C** Lipids (fats and oils) are present
- D** Proteins are present

A child eating only ugali develops a swollen belly (oedema). Which explanation is scientifically correct?

- A** The body has no keratin, so the belly cannot hold its shape.
- B** The child has no DNA to carry the genetic code.
- C** The body cannot make enough albumin, a protein, because too few amino acids come from food.
- D** Too much starch is stored in the belly as fat.

A child in Kisumu eats githeri (maize and beans) regularly and stays healthy. What is the best chemical explanation?

- A** Beans contain more starch, giving the child more energy than ugali does.
- B** Maize in githeri contains protein instead of starch.
- C** Githeri contains more water than ugali, so the child stays hydrated.
- D** Maize supplies carbohydrate and beans supply protein, providing the amino acids the body needs for protein synthesis.

During the Driving Question Board review, a learner asks: "How do cells use DNA to know which proteins to build?" How should the class sort this question?

- A** Not a science question, so it should be removed
- B** Answered, because githeri contains protein
- C** New Question Generated, to investigate in a future unit on genetics
- D** Answered, because amino acids explain kwashiorkor

A typical Kenyan daily menu is: breakfast, uji (thin maize porridge) with sugar; lunch, ugali with sukuma wiki; supper, rice with a small piece of fried fish. Which nutritional gap is most likely?

- A** No carbohydrates, because uji is thin
- B** No lipids, because fried food contains no fats or oils
- C** Insufficient protein, because the only protein source is a small piece of fish
- D** Too much protein, because fish is eaten

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).